

Lecture 2: Data Representation I — Bits, Bytes & Integers

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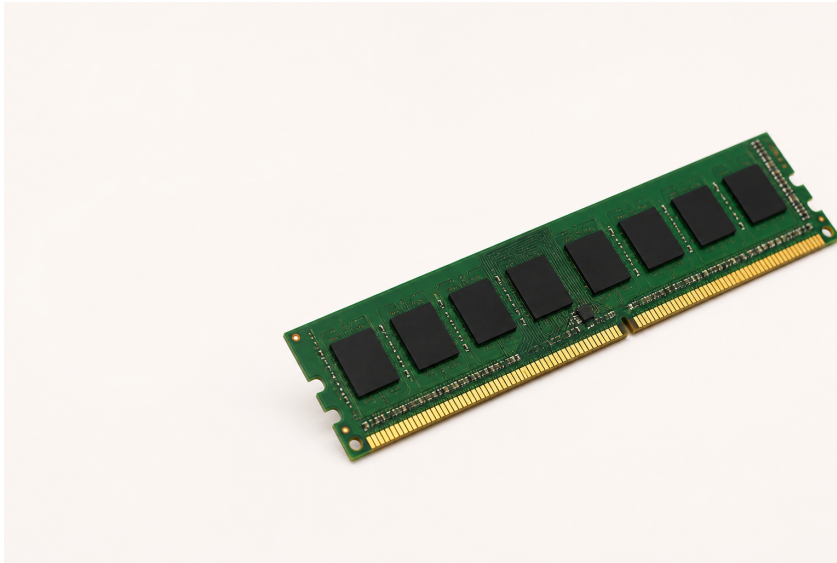
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From Physical Memory to One Byte

That is a **RAM module** — physical hardware, many chips, not one byte.

Memory is **byte-addressable**: every address names one **8-bit byte**, and a byte is the smallest unit a C program can address on its own.

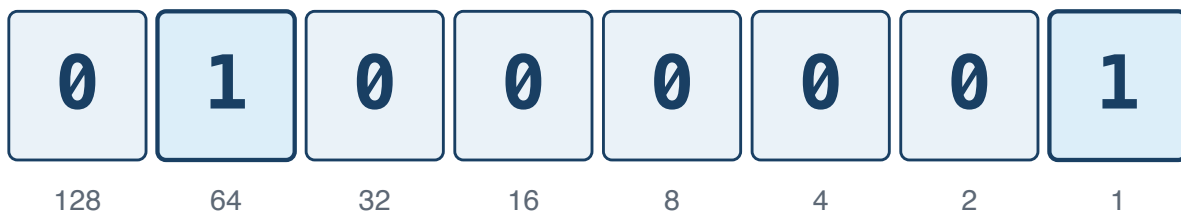


One byte = 8 bits. Today is about what fits in one, and how to work on it.

One Byte, Two Readings

Start with one byte: eight switches that are each either 0 or 1.

One byte: the bit pattern for 65



$$64 + 1 = 65$$

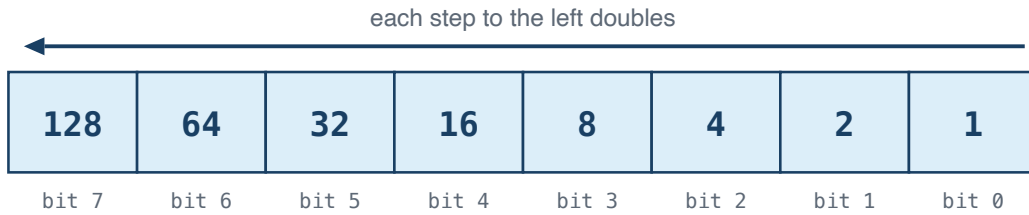
as text (ASCII): A

The computer stores only the bits; the program decides how to read them.

Counting in Binary

Why does **01000001** mean 65? Because each place is worth twice the one to its right — the same idea as decimal's ones, tens and hundreds, with 2 instead of 10.

Each place is worth twice the place to its right



Turn a bit on and you add its place value.

$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$

Counting Up

Counting works exactly the way it does in decimal — when a column runs out of digits, it rolls over and carries into the next one:

0 = 0000	3 = 0011	6 = 0110	9 = 1001
1 = 0001	4 = 0100	7 = 0111	10 = 1010
2 = 0010	5 = 0101	8 = 1000	11 = 1011

Each extra bit doubles the patterns available: 8 bits give $2^8 = 256$. So an unsigned byte runs 0–255, and an N-bit unsigned value runs 0 to $2^N - 1$ — its **UMax**, modulo 2^N .

Writing the Same Number Three Ways

A number does not change when you write it differently. C lets you say which notation you are using with a **prefix**:

Written	Prefix means	Value
42	no prefix — plain decimal	forty-two
0b101010	0b — read the digits as binary	forty-two
0x2A	0x — read the digits as hexadecimal	forty-two

All three are the same value to the compiler. You pick whichever makes the thing you are doing readable: decimal for counting, binary for seeing bits, hex for writing bytes compactly.

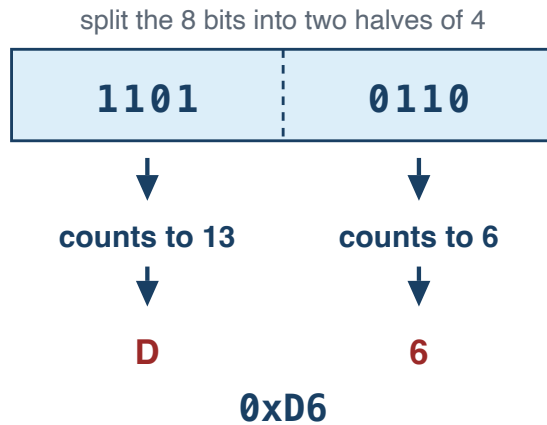
Note

0b is not standard until C23. GCC and Clang accept it, and these slides use it to show bit patterns, but in ISO C17 write **0x2A** or **42**.

Why Hex Fits Bytes So Well

Eight ones and zeros is easy to miscount. Four bits count 0–15 — exactly one hex digit — so one byte is always two digits, and you can still read the bits straight off the page.

One byte = two hex digits



**4 bits count 0 to 15,
so we need 6 extra digits:**

10 = A
11 = B
12 = C
13 = D
14 = E
15 = F

0 through 9 stay as they are.

Characters Are Just Numbers

The rule that turns 65 into A is **ASCII** — the American Standard Code for Information Interchange — a table agreed in the 1960s that gives every character a number from 0 to 127.

Characters	Codes	Worth knowing
'0'–'9'	48–57 (0x30–0x39)	consecutive, so <code>c - '0'</code> gives the digit
'A'–'Z'	65–90 (0x41–0x5A)	consecutive
'a'–'z'	97–122 (0x61–0x7A)	exactly 32 more than the capitals

```
char c = '7';
int d = c - '0';           // 7
int is_digit = (c >= '0' && c <= '9'); // 1
```

In C the literal 'A' is the number 65. Nothing clever is happening.

The Whole Table

ASCII in full — every code from 0 to 127

Read down a column. Grey are control codes with no glyph; the tinted blocks are the digits, the capitals and the lowercase letters.

0	00	NUL	16	10	DLE	32	20	SP	48	30	0	64	40	@	80	50	P	96	60	`	112	70	p
1	01	SOH	17	11	DC1	33	21	!	49	31	1	65	41	A	81	51	Q	97	61	a	113	71	q
2	02	STX	18	12	DC2	34	22	"	50	32	2	66	42	B	82	52	R	98	62	b	114	72	r
3	03	ETX	19	13	DC3	35	23	#	51	33	3	67	43	C	83	53	S	99	63	c	115	73	s
4	04	EOT	20	14	DC4	36	24	\$	52	34	4	68	44	D	84	54	T	100	64	d	116	74	t
5	05	ENQ	21	15	NAK	37	25	%	53	35	5	69	45	E	85	55	U	101	65	e	117	75	u
6	06	ACK	22	16	SYN	38	26	&	54	36	6	70	46	F	86	56	V	102	66	f	118	76	v
7	07	BEL	23	17	ETB	39	27	'	55	37	7	71	47	G	87	57	W	103	67	g	119	77	w
8	08	BS	24	18	CAN	40	28	(	56	38	8	72	48	H	88	58	X	104	68	h	120	78	x
9	09	HT	25	19	EM	41	29	)	57	39	9	73	49	I	89	59	Y	105	69	i	121	79	y
10	0A	LF	26	1A	SUB	42	2A	*	58	3A	:	74	4A	J	90	5A	Z	106	6A	j	122	7A	z
11	0B	VT	27	1B	ESC	43	2B	+	59	3B	;	75	4B	K	91	5B	[	107	6B	k	123	7B	{
12	0C	FF	28	1C	FS	44	2C	,	60	3C	<	76	4C	L	92	5C	\	108	6C	l	124	7C	
13	0D	CR	29	1D	GS	45	2D	-	61	3D	=	77	4D	M	93	5D	]	109	6D	m	125	7D	}
14	0E	SO	30	1E	RS	46	2E	.	62	3E	>	78	4E	N	94	5E	^	110	6E	n	126	7E	~
15	0F	SI	31	1F	US	47	2F	/	63	3F	?	79	4F	O	95	5F	_	111	6F	o	127	7F	DEL

Each cell is decimal, hex, then the character. 'a' - 'A' = 32 — one bit apart.

And Above 127?

128 to 255: not ASCII, and not one answer

These are the ISO 8859-1 (Latin-1) meanings. Windows-1252 disagrees about 128–159.

In UTF-8 — what your files almost certainly are — none of these bytes is a character on its own.

128 80 ctl	144 90 ctl	160 A0	176 B0 °	192 C0 Å	208 D0 Đ	224 E0 à	240 F0 ð
129 81 ctl	145 91 ctl	161 A1 ¡	177 B1 ±	193 C1 Á	209 D1 Ñ	225 E1 á	241 F1 ñ
130 82 ctl	146 92 ctl	162 A2 ¢	178 B2 ²	194 C2 Â	210 D2 Ò	226 E2 â	242 F2 ò
131 83 ctl	147 93 ctl	163 A3 £	179 B3 ³	195 C3 Ã	211 D3 Ó	227 E3 ã	243 F3 ó
132 84 ctl	148 94 ctl	164 A4 ¤	180 B4 ´	196 C4 Ä	212 D4 Ô	228 E4 ä	244 F4 ô
133 85 ctl	149 95 ctl	165 A5 ¥	181 B5 µ	197 C5 Å	213 D5 Õ	229 E5 å	245 F5 õ
134 86 ctl	150 96 ctl	166 A6 ¦	182 B6 ¶	198 C6 Æ	214 D6 Ö	230 E6 æ	246 F6 ö
135 87 ctl	151 97 ctl	167 A7 §	183 B7 ·	199 C7 Ç	215 D7 ×	231 E7 ç	247 F7 ÷
136 88 ctl	152 98 ctl	168 A8 ¨	184 B8 ¨	200 C8 È	216 D8 Ø	232 E8 è	248 F8 ø
137 89 ctl	153 99 ctl	169 A9 ©	185 B9 ¹	201 C9 É	217 D9 Ù	233 E9 é	249 F9 ù
138 8A ctl	154 9A ctl	170 AA ¢	186 BA ¢	202 CA Ê	218 DA Ú	234 EA ê	250 FA ú
139 8B ctl	155 9B ctl	171 AB «	187 BB »	203 CB Ë	219 DB Û	235 EB ë	251 FB û
140 8C ctl	156 9C ctl	172 AC ¬	188 BC ¼	204 CC Ì	220 DC Ü	236 EC ì	252 FC ü
141 8D ctl	157 9D ctl	173 AD	189 BD ½	205 CD Í	221 DD Ý	237 ED í	253 FD ý
142 8E ctl	158 9E ctl	174 AE ®	190 BE ¾	206 CE Î	222 DE Þ	238 EE î	254 FE þ
143 8F ctl	159 9F ctl	175 AF ¯	191 BF ¿	207 CF Ï	223 DF ß	239 EF ï	255 FF ÿ

A byte above 127 means nothing until you know the encoding it was written in.

Ñ is 209 here and ñ is 241 — but in UTF-8 each one takes **two** bytes, so "Peña" is 5 bytes, not 4.

ASCII stops at 127 — it only ever needed 7 bits. Everything above that is a *different standard's* idea of what the byte means.

So — Can We Type Spanish?

Put the two tables together and you have 256 slots. Is that enough for every character Spanish actually needs?

Every character Spanish needs has a slot

Blue came from the ASCII table. Orange came from the half above 127.

¿	C	u	á	n	t	o	s	_	a	ñ	o	s	?
191	67	117	225	110	116	111	115	32	97	241	111	115	63

14 characters, 14 bytes — one byte each, and nothing is missing.

ñ Ñ á é í ó ú Á É Í Ó Ü ü ¿ ¸ — all sixteen live in the upper half.

Important

Yes — but only because someone chose Latin-1. That same byte 241 is á in Latin-2, я in KOI8-R and ϱ in Greek, and nothing in the file says which. Spanish and Polish cannot share one 8-bit document at all. That is what UTF-8 fixed, at the price that ñ takes two bytes — so `strlen("Peña")` is 5.

How Does C Know What to Print?

The bits never say what they are. `printf` prints whatever the **format code** asks for — so the same `x` comes out as a number or as a character:

```
int x = 65;
printf("%d\n", x);    // 65
printf("%c\n", x);    // A

unsigned a = 12, b = 10;
printf("%u\n", a & b); // 8
printf("%u\n", a << 2); // 48
```

Nothing about `x` changed between those first two lines. The only thing that changed is what you told `printf` to do with it.

Telling `printf` What You Handed It

`printf` cannot see the type of what you pass — the format string is how it finds out. Get that wrong and the output is garbage, or worse:

Code	Prints
<code>%d</code>	a signed int
<code>%u</code>	an unsigned
<code>%c</code>	one character
<code>%s</code>	a string (<code>char *</code>)
<code>%zu</code>	a <code>size_t</code> , as from <code>sizeof</code>

Code	Prints
<code>%x %X</code>	hex, lower or upper case
<code>%f</code>	a double
<code>%p</code>	a pointer — cast it to <code>void *</code>
<code>%%</code>	a literal <code>%</code> sign
<code>%02X</code>	hex, padded to 2 with zeros

Warning

Build with `-Wall` and the compiler catches most mismatches — `%s` with an `int`, `%d` with a `double`. It stays quiet about `%d` with an `unsigned`, though, because they are the same width. That one is still wrong, and it is the one you will write.

Bit-Level Operations

For these bit patterns, C provides six bitwise operators:

Op	Name	Example
<code>&</code>	AND	<code>0b1100 & 0b1010 = 0b1000</code>
<code>\ </code>	OR	<code>0b1100 \ 0b1010 = 0b1110</code>
<code>^</code>	XOR(Exclusive OR)	<code>0b1100 ^ 0b1010 = 0b0110</code>
<code>~</code>	NOT	<code>~0b1100 = 0b1111...11110011</code> (every bit flips, including the leading zeros)
<code><<</code>	Left shift	<code>0b0011 << 2 = 0b1100</code>
<code>>></code>	Right shift	<code>0b1100 >> 2 = 0b0011</code>

Warning

Right shift of signed integers is **implementation-defined** in C. GCC uses **arithmetic** right shift (preserves sign bit). Don't rely on this unless you're sure of your compiler.

Bitwise Operations, One Column at a Time**The same two inputs, three separate rules**

Every output bit is decided on its own, from just the two bits directly above it.

INPUTS						
a	1	1	0	0	= 1100	
b	1	0	1	0	= 1010	
a & b	1	0	0	0	AND = 1000	1 only where both are 1
a b	1	1	1	0	OR = 1110	1 where either is 1
a ^ b	0	1	1	0	XOR = 0110	1 where they differ

De Morgan's Laws

Two rules let you rewrite any mix of &, | and ~ — and they are how Lab 1 asks you to build ^ out of nothing but ~ and &.

Flip the whole thing, and AND becomes OR

$$\sim(x \ \& \ y) == \sim x \ | \ \sim y$$

$$\sim(x \ | \ y) == \sim x \ \& \ \sim y$$

x	y	x & y	$\sim(x \ \& \ y)$	$\sim x \ \ \sim y$
0	0	0	1	1
0	1	0	1	1
1	0	0	1	1
1	1	1	0	0

same in all four rows

Two inputs give four possible rows, so checking all four is a complete proof.

Shifting Bits

Shifting slides every bit sideways

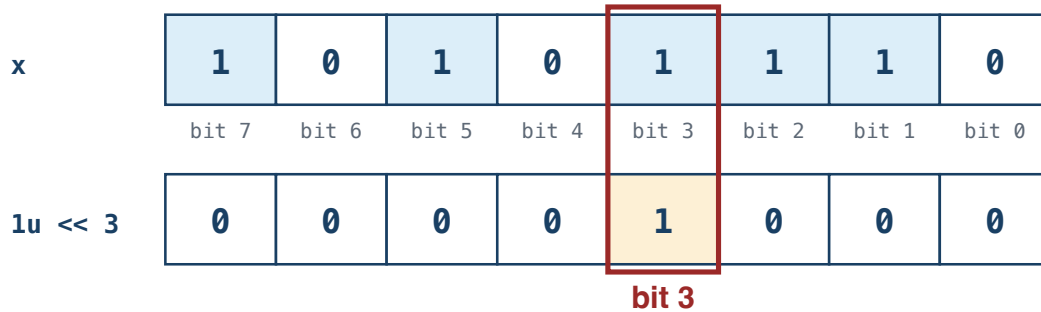
x	0	0	0	0	1	1	0	0	= 12
	←								
x << 2	0	0	1	1	0	0	0	0	= 48 (× 4)
	→								
x >> 2	0	0	0	0	0	0	1	1	= 3 (÷ 4)

Bits pushed off the end are gone for good; zeros come in behind them.

Shifting left by n multiplies by 2^n . Shifting an unsigned value right by n divides by 2^n .

Test, Set, Clear, Toggle

$1u \ll 3$ builds a mask with exactly one bit set



Bit 0 is the rightmost — the ones place. The mask is 0 everywhere except the bit you want.

```

unsigned x = 0xAE;           // 10101110
x |= (1u << 3);              // SET    - bit 3 becomes 1
x &= ~(1u << 3);             // CLEAR - bit 3 becomes 0
x ^= (1u << 3);              // TOGGLE - bit 3 flips
if (x & (1u << 3)) { }      // TEST  - nonzero when bit 3 is 1
  
```

Those four are most of Lab 1. Keep the value **unsigned** — shifting into a sign bit is undefined.

What Two's Complement Means

Complement means the part that completes a whole. Flip every bit of x to get its *one's complement*; add 1 to that and you have its **two's complement**, which is how C writes $-x$:

To write -5 in 8 bits

1. Start with +5

00000101

↓ flip every 0 ↔ 1

2. One's complement

11111010

+ 1

↓

3. Two's complement

11111011

= -5

More 8-bit examples

+1: 00000001

-1: 11111111

0: 00000000

two's: 00000000

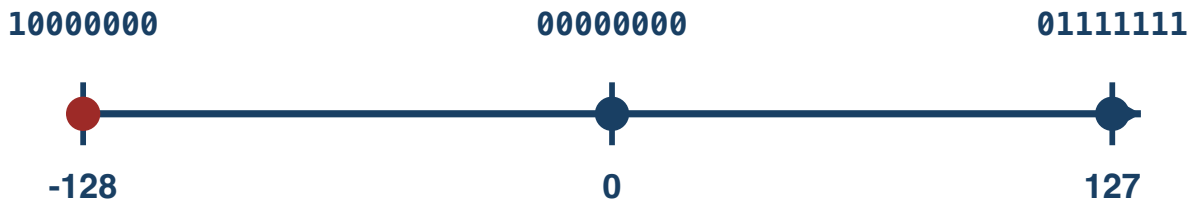
Zero stays zero.

A ninth bit is simply discarded — 8 bits keep only their rightmost 8.

The Range It Buys You

An N -bit two's complement covers $-2^{(N-1)}$ to $2^{(N-1)} - 1$:

8-bit two's-complement values



There is one more negative value than positive value.

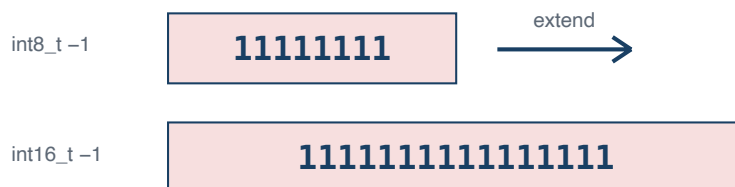
$T_{\min} = -T_{\max} - 1$ — there is no positive 128 in 8 bits, so negating $T_{\min}$ overflows and is undefined.

Sign Extension

When converting a smaller signed type to a larger one, copy the sign bit:

```
int8_t  x = -1;    // 0b11111111
int16_t y = x;     // 0b1111111111111111 = -1 ✓
```

Copy the sign bit into the new high bits

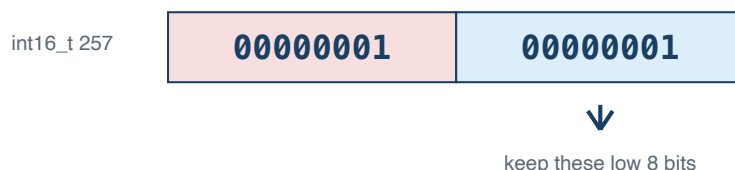


Truncation

When converting larger to smaller, drop the high bits:

```
int16_t x = 257;
uint8_t y = x;    // 1: conversion to unsigned is modulo 256
```

Keep the low bits; discard the high bits



`int8_t result = 00000001 = 1`

Size of Integer Types

```
printf("char:      %zu\n", sizeof(char));    // 1 C byte (not always 8 bits)
printf("short:     %zu\n", sizeof(short));   // 2
printf("int:       %zu\n", sizeof(int));     // 4
```

```
printf("long:      %zu\n", sizeof(long));    // 8 (LP64)
printf("long long: %zu\n", sizeof(long long)); // 8
printf("pointer:   %zu\n", sizeof(void*));   // 8
```

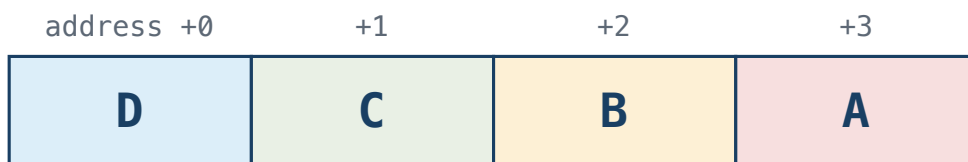
Tip

Use `sizeof` and fixed-width types (`int32_t`, `uint64_t`) when exact sizes matter. Never assume `int` is 4 bytes; use `CHAR_BIT` when bit width matters. Plain `char` is its own trap: whether it is signed is implementation-defined, so say `signed char` or `unsigned char` when it matters.

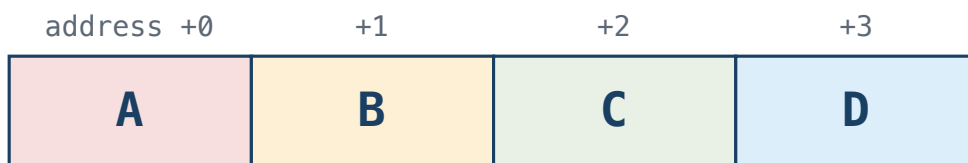
Byte Ordering (Endianness)

An `int` occupies 4 consecutive bytes, so those 4 bytes have to be laid out in some order. **Endianness** is that order. One question matters: **which byte goes at the lowest address?**

Little-endian: D is first in memory



Big-endian: A is first in memory



```
int x = 1;    // ask the machine: which byte of x sits at the lowest address?
if (*(unsigned char *)&x == 1) printf("Little-endian\n");
else                             printf("Big-endian\n");
```

Bit Tricks in the Wild

LeetCode 136 — Single Number. Every value appears twice except one. Find it in one pass, using no extra memory.

```
int singleNumber(int *nums, int n) {
    int x = 0;
    for (int i = 0; i < n; i++)
        x ^= nums[i];
    return x;
}
```

nums: 5 5 1 2 2

```
x = 0
^ 5 -> 5
^ 5 -> 0   the pair cancels
^ 1 -> 1
^ 2 -> 3
^ 2 -> 1   answer
```

$a \wedge a$ is 0, $a \wedge 0$ is a , and XOR does not care about order — so every pair annihilates wherever the two copies sit, and the unpaired value is what survives. Shuffle the array to 2 5 1 5 2 and the answer is still 1.

AI Systems Connection

Quantization is one important reason large models can fit on more hardware. A common symmetric int8 scheme represents a block of weights with 8-bit signed values and a scale factor:

$$\text{real_value} \approx \text{scale} \times \text{int8_value} \qquad \text{scale} = \text{max_abs} / 127$$

Everything today shows up here. Two int8 values need 16 bits for their product, so a long dot product accumulates into something wider — often `int32_t`. And if a signed weight is loaded as *unsigned* into a wider register, sign extension does not happen, negatives become huge positives, and the answer is quietly wrong.

Try it: quantize a float array with `scale = max_abs / 127`, dequantize, and measure the error.

Practice

1. Write `0xD6` in binary, and `10110010` in hex, without using a computer.
2. What does `x & 1` tell you about `x`? And what does `x & (x - 1)` do? Try both on `0b0110` by hand.
3. Write one expression that clears bit 5 of `unsigned x`, and one that tests whether bit 0 and bit 7 are both set.
4. What is `TMax` for a 32-bit `int`? What is `TMin`?
5. Write a C expression that checks if your machine is little-endian.
6. Is left-shifting a 32-bit `int` by 32 positions defined in C? Why not?
7. Why is `-TMin` not representable in the same signed type?

Reading

- CS:APP §2.1 (Information Storage)
- CS:APP §2.2 (Integer Representations)